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Ivy-like electrospun block copolymer fibers supported polymer solid electrolytes with wide electrochemical stability window for all-solid-state Li-metal battery

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ABSTRACT

All-solid-state lithium metal batteries (ASSLMBs) are considered as next-generation energy storage devices due to their high specific energy density and safety, and solid polymer electrolytes (SPEs) are an important component of ASSLMBs. Poly(ethylene oxide) (PEO) is the most prevalent matrix of SPE but is criticized for its poor mechanical properties and a narrow electrochemical stability window. To address this issue, poly(vinylidene fluoride)-*b*-poly(poly(ethylene glycol) methyl ether methacrylate (PVDF-*b*-PPEGMA) copolymer fibers with an ivy-like morphology were prepared by electrospinning and used to support PEO/lithium salt electrolytes. The electrospun PVDF-*b*-PPEGMA copolymer fiber membrane provides an uncial hierarchical structure with large pore size, good compatibility and adhesion with PEO due to side ethylene oxide units in the PPEGMA segment. The ivy-like fibers supported SPE exhibited good thermal stability and high mechanical strength. Furthermore, ASSLMBs assembled using the above SPE, $LiFePO_4$ cathode, and lithium metal anode possessed a wide electrochemical stable window (5.58 V vs. Li/Li^+) and good initial discharge capabilities at 60 °C.

1. Introduction

Lithium ion batteries (LIBs) are considered as the most promising energy storage device for portable electronic devices, electric vehicles and large-scale energy storage systems, due to their high energy density, long cycle life and low cost [1–4]. However, the energy density of current LIBs cannot meet demands of the next generation of energy storage devices. Using of metallic lithium as an anode material is a feasible way to improve the specific energy density [5–7]. However, lithium metal is highly active and prone to forming dendrites. These dendrites tend to progressively expand and potentially penetrate the separator during their growth process, which can further lead to the leakage of liquid electrolytes and potential safety hazards [8–12].

The electrolyte is an important component of lithium batteries, acting as an ion-conducting medium and a separator. The development of all-solid electrolytes, which completely eliminate liquid components, is an important way to improve the safety of lithium batteries [13]. All-solid electrolytes can be divided into the solid polymer electrolytes (SPEs) and inorganic solid electrolytes. Inorganic electrolytes exhibit high ionic conductivities and good electrochemical stability, but encounter challenges of interfacial impedance and poor inhibition of lithium dendrite formation. SPEs have sparked great interest among researchers due to their superior thermal stability, mechanical strength and interfacial compatibility with electrodes [14–18]. So far, various polymers have been investigated as matrices of SPEs.

Poly(ethylene oxide) (PEO) is a prevalent polymer electrolyte matrix due to a complexation between ethylene oxide (EO) units and lithium ions, but PEO exhibits disadvantages of high crystallinity, poor film-forming ability and limited mechanical strength. Furthermore, PEO cannot tolerate a high-voltage over 4 V, which limits its compatibility with high-voltage cathode materials [19]. Consequently, modification of PEO is frequently essential to enhance its comprehensive performance. Polymers based on vinylidene fluoride (VDF) monomers are another kind of host for SPEs due to their excellent electrochemical and mechanical properties, excellent chemical and thermal stabilities, and wide voltage windows [20]. The high electronegativity of fluorine atoms in VDF (co)polymers enables them to complex with lithium ions and endows them with a certain ion-conducting ability [21,22]. However, VDF

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(co)polymers exhibit low conductivities at ambient temperature (10^{-9} S/cm) and only show the application prospect in the gel form [23–25]. The gel electrolyte, a quasi-solid-state electrolyte typically obtained by incorporating a small amount of organic solvents or liquid plasticizer within a polymer host, still fails to address safety and mechanical strength requirements, due to the retention of liquid components [13].

The blending of PEO and VDF (co)polymer is always adopted to obtain polymer electrolytes combined with the advantages of two components. In addition to common melting and solution blending, incorporating of one-dimensional (1D) nanomaterials (structural engineering) can facilitate the electrons/ions transport along their longitudinal axis and thus improve the performance of electrolytes [26,27]. Electrospinning is an effective strategy to prepare 1D VDF (co)polymer fibers with improved β crystalline and piezoelectricity, endowing their potential applications in sensors [28], self-powered generators [29], energy harvesters [30], wearable devices [31] and functional coatings [32]. The application of electrospun VDF (co)polymer fibers (or membranes) in solid-state electrolytes also garnered significant attentions. **Wang** *et al* found that electrospun PVDF fibers supported polymer electrolytes possessed enhanced mechanical strength and high-voltage operation up to 4.5 V [27]. Zhou *et al* prepared a composite electrolyte with electrospun PVDF fiber and determined the transference number of 0.37 [33]. Choi *et al* employed eletrospun PVDF nanofiber webs as an electrolyte binder or a separator for a battery [34]. These studies demonstrate the incorporation of electrospun PVDF fibers (or membranes) in solid-state electrolytes would suppress the growth of lithium dendrite and improve the mechanical strength and ionic conductivity, evidencing the potential application of electrospun PVDF fibers (or membranes) in the field of solid-state electrolytes.

Ivy is a root-climbing liana with the remarkable ability to anchor plants and objects due to its unique attachment strategy [35]. Inspired by this characteristic, nanofiber membranes with ivy-like morphology and enhanced adhesive forces were prepared by electrospinning of poly (vinylidene fluoride)-*b*-poly(poly(ethylene glycol) methyl ether methacrylate) (PVDF-*b*-PPEGMA) copolymers and composited with PEO/lithium salt to fabricate SPEs in the present work. The hierarchical morphology of the electrospun PVDF-*b*-PPEGMA copolymer membrane ensures both enhancements of mechanical strength and permeation of the PEO matrix into the internal framework of the membrane. Meanwhile, side EO units in PPEGMA segments act as the anchoring structure in ivy, firmly gripping the PEO matrix. Hence, the lithium metal battery assembled using SPEs composed of PEO and ivy-like nanofiber membranes showed outstanding electrochemical performances, indicating the potential application of this novel SPE in high-performance ASSLMBs.

2. Experimental section

2.1. Materials

Poly(vinylidene fluoride) with iodine end group (PVDF-I) and a number average molecular weight (Mn) of 65,000 was prepared through iodine transfer polymerization (ITP) of VDF. Poly(ethylene glycol) methyl ether methacrylate (PEGMA) with Mn of 300, PEO with a viscosity average molecular weight (Mv) of 100,000, and bis(trifluoromethane sulfonimide) (LiTFSi) with purity > 99 % were purchased from Sigma Aldrich Inc. 2,2′-azobis-(2,4-dimethylvaleronitrile) (ABVN), dimethyl sulfoxide-d_6 (d_6-DMSO), 2,2′-azobis-2-methylbutyronitrile and ethyl alcohol were purchased from Sinopharm Chemical Reagent Co., Ltd. *N*, *N*-dimethylacetamide (DMF), acetonitrile and *N*-methyl-2-pyrrolidinone (NMP) were purchased from Aladdin Inc. $LiFePO_4$, conductive carbon black (Supper-P) and PVDF (Kynar HSV900) were purchased from Shenzhen Kejing Star Technology Co., Ltd.

2.2. Synthesis of PVDF-b-PPEGMA copolymers

PVDF-*b*-PPEGMA copolymers were synthesized via ITP of PEGMA using PVDF-I as a macro-iodine transfer agent. PVDF-I powder, PEGMA, ABVN and dry NMP were added to a glass reactor and dissolved under the stirring. The copolymerization was conducted at 50 °C for 12 h under stirring and nitrogen flow. The copolymer was obtained after precipitating the reactant with cold methanol, repeated washing with water and drying. PVDF-*b*-PPEGMA copolymers with different block ratios were synthesized by varying the feed mass ratios between PVDF-I and PEGMA.

2.3. Solution electrospinning of PVDF and PVDF-b-PPEGMA copolymers

PVDF-*b*-PPEGMA copolymer nanofiber membranes were fabricated via the solution electrospinning method. The dried PVDF-*b*-PPEGMA copolymer was dissolved into DMF with a mass concentration of 20 % by magnetic stirring at 40 °C. The resulting homogeneous solution was loaded into a pipette and fed through a high-pressure electrospinning apparatus. Electrospinning conditions were designed as follows: voltage 16 ~ 19 kV (increased as the molecular weight of copolymer decreased), tip-to-collector distance (TTDC) 15 cm, solution feed rate 1 ml/h^{-1}, collector drum rotation speed 360 rpm, and humidity < 70 %. The copolymer nanofibers were collected at a grounded, aluminum drum wrapped with commercial silicon paper. The collected copolymer nanofiber membranes were vacuum-dried at 60 °C for 12 h to remove residual solvents. For comparison, the same steps were applied to prepare an electrospun VDF homopolymer nanofiber membrane.

2.4. Fabrication of solid polymer electrolytes and all-solid-state Li-metal batteries

PEO and LiTFSi salt were previously vacuum dried at 60 °C and 120 °C to remove any moisture, respectively. PEO was dissolved in acetonitrile with a mass concentration of 2 % by vigorously magnetic stirring until completely dissolved. To obtain an optimized SPE, LiTFSi salt was added into the PEO solution with designed EO/Li ratios. The as-prepared electrospun copolymer membrane was soaked in PEO/LiTFSi solution for 5 min, and then dried under vacuum at 60 °C for 12 h to remove the residual solvent, followed by heat-pressing at 10 MPa and 120 ℃. The same process was repeated to prepare SPEs using a PVDF electrospun membrane. SPE membranes containing electrospun PVDF-*b*-PPEGMA copolymer nanofibers and PVDF homopolymer nanofibers were named i-SPE and n-SPE, and i-SPE and n-SPE with EO/Li ratios of 4/1, 6/1, 8/1, 10/1, 12/1, 14/1 and 16/1 were added the suffix 1, 2, 3, 4, 5, 6 and 7, respectively. The i-SPE and n-SPE membranes were cut into circles with a diameter of 19 mm, and a thickness of 70 ± 5 μm and 10 ± 5 μm, respectively. All SPE pieces were stored in a glove box before use.

$LiFePO_4$ cathodes were prepared by mixing commercial $LiFePO_4$, conductive carbon black and PVDF with a mass ratio of 8:1:1, dissolving the mixture in NMP solvent to form a uniform slurry, and casting the slurry onto an aluminum foil, drying under vacuum at 120 °C and punching into round electrodes with 14 mm diameter. The CR 2025 button batteries were assembled as $LiFePO_4$/SPE/Li cells in an argon − filled glove box.

2.5. Characterization of block copolymers, electrospun membranes, SPEs and Li-metal battery

The structure of PEGMA and PVDF-*b*-PPEGMA copolymers were determined by a 500 MHz NMR spectrometer (Bruker Co., Ltd. Germany) using d_6-DMSO as the solvent at room temperature.

A field emission scanning electron microscopy (FE-SEM) (SU-8010, HITACH Co., Ltd. Japan) was applied to observe morphological characteristics of electrospun PVDF-*b*-PPEGMA copolymer membranes and SPEs. Dry and flat membranes were adhered to the conducting tape and

sprayed with gold for 60 s before testing. Wide angle X-ray scattering (WAXS) analysis of SPE was conducted on the beamline of a nano-insider vertical WAXS/SAXS system (Senocs, Co, Ltd. France). The wavelength of the X-ray radiation was 0.124 nm. Scattering patterns were recorded by a Pilatus 100 k detector (Dectris, Swiss).

Differential scanning calorimetry (DSC) measurements were carried out using NETZSCH 214 (DSC, NETZSCH, Germany) equipped with an IC70 cooler. The thermal protocol used for non-isothermal crystallization and melting was as follows: (1) cooling to −70 °C at a rate of 10 °C·min^{-1}; (2) heating to 200 °C at the rate of 10 °C min^{-1} to eliminate heat history;(3) cooling to −70 °C at the rate 10 °C min^{-1} again to analyze crystallization behavior; (4) heating to 200 °C at the rate of 10 °C min^{-1} to analyze melting behavior. Thermogravimetric analysis (TGA) (Mettler TGA/DSC3+, Switzerland) was performed to analyze thermal stability of SPEs under nitrogen atmosphere at a heating rate of 10 °C min^{-1} from 50 °C to 600 °C.

The tensile properties of electrospun membranes and SPEs were measured using a universal testing machine (SUNS UTM4203, China) equipped with a temperature control unit. Membranes were cut into a dumbbell shape with a neck length of 15 mm, a neck width of 3 mm and a thickness of 0.06 ~ 0.08 mm. The test was conducted at a speed of 5 mm min^{-1}.

The sandwich core cell of SPEs between two stainless electrodes was assembled to test ionic conductivity, which was performed with an alternating current (AC) impedance method at an amplitude of 5 mV over the frequency range of 0.01 ~ 1000 kHz Diameters of stainless-steel electrode and SPE are 16 mm and 19 mm separately. The ionic conductivity σ (S cm^{-1}) of the electrolyte was calculated using Equation (1):

$$\sigma = L/AR_b \tag{1}$$

where L (cm) is the thickness of SPE determined by a digital micrometer, A the electrode area (cm^2), and R_b the bulk resistance value determined by the intercept of the low-frequency signal with the real axis in the Nyquist plot of EIS.

The lithium ion transference number (t^+) of a symmetric lithium battery was measured by a combination of AC impendence (0.01 ~ 1000 kHz) and direct current (DC) polarization measurement, and calculated using Equation (2):

$$t^+ = \frac{I_{ss}(\Delta V - I_0R_0)}{I_0(\Delta V - I_{ss}R_{ss})} \tag{2}$$

A small 10 mV potential (ΔV) was normally applied across a lithium metal symmetric cell. I_0 and I_{ss} are the initial current value and a stable value of current according to DC curve. R_0 and R_{ss} are values of bulk resistance before and after polarization, which can be obtained as the same as R_b.

Linear sweep voltammetry (LSV) was performed on an asymmetrical stainless steel(working electrode)/SPE/Li (counter and reference electrode) cell with scan rate of 1 mV s^{-1}. The test was conducted between 0 and 6 V to examine electrochemical stability of SPE. The interfacial stability of SPE against lithium metal was examined by recording the AC impedance spectrum of a symmetric Li/i-SPE-6/Li cell over storage time at 60 °C.

Cyclic voltammetry (CV) characteristics of $LiFePO_4$/SPE/Li sandwich cells were measured at a scanning speed of 1 mV s^{-1} from 2.5 to 4.2 V at 60 °C on a CHI660E electrochemical workstation (Shanghai Chenhua Instrument Co, Ltd.). The galvanostatic charging-discharging and rate capability of cells were carried out in the NEWARE System (CT-4008) (1C = 170 mAh g^{-1}). Long cycling charge and discharge test of the solid-state LFP/i-SPE6/Li battery was conducted at a constant current rate in the voltage range of 2.5 ~ 4.0 V. The rate measurement was conducted at 0.1, 0.2, 0.5,1 and 2C with five cycles at each rate under 60 °C.

3. Results and discussion

3.1. Morphology of electrospun copolymer and SPE membranes

PVDF-*b*-PPEGMA copolymers with different compositions were synthesized by ITP of PEGMA mediated by PVDF-I macromolecular chain transfer agent. H^1 NMR spectrum of PEGMA, PVDF-I and typical PVDF-*b*-PPEGMA copolymer is shown in Fig. S1, and the block ratio between PVDF to PPEGMA can be determined from peak areas of corresponding hydrogen atoms of PVDF and PPEGMA. Thus, copolymers with PVDF/PPEGMA block ratios of 100/4.7, 100/7.3 and 100/10 were applied in the electrospinning process to fabricate porous scaffold membranes. SEM images of electrospun PVDF-*b*-PPEGMA copolymer fibers with different PVDF/PPEGMA block ratios are shown in Fig. 1.

It can be seen that hierarchical structured nanofibers with an extremely nonuniform distribution in diameter and large pore sizes are formed in all electrospun block copolymer membranes. As shown in Fig. 1a, the fibers contained a lot of beads for the copolymer with a PVDF/PPEGMA block ratio of 100/4.7. A higher applied voltage is needed in the electrospinning process of PVDF-*b*-PPEGMA copolymers with a low block ratio (and low molecular weight) and the fibers becomes thinner [36]. As the PVDF/PPEGMA block ratio increased to 100/7.3, beads rapidly decreased and multi-level nanofiber structure became distinctive (Fig. 1b). As the PVDF/PPEGMA block ratio increased to 100/10, fibers were smooth without any beads, and an ivy-like morphology was built up composed of trunk fibers with a diameter of 500 nm to 1 μm and branch fibers with a diameter of 10 nm ~ 100 nm (Fig. 1c). Although the formation of the ivy-like structure electrospun fibers is more perfected to PVDF-*b*-PPEGMA copolymers with high molecular weights (chain lengths) of PPEGMA block, the preparation of block copolymers with long PPEGMA segments tends to be difficult since PEGMA oligomer exhibits a great steric hindrance in the chain extension polymerization of PEGMA.

The configuration of the PVDF-*b*-PPEGMA copolymer resembles a "tadpole" more than a long chain. Due to the influence of the tadpole-like configuration and the different conductivities between PPEGMA and PVDF segments, the electrostatic field force was unevenly distributed during the electrospinning process. Fig. 2a illustrates the relationship between electrostatic field forces (strength and distribution) and steric hindrance for PVDF-*b*-PPEGMA copolymers with different molecular weights. The charge jet split into fibers with different diameters due to an uneven electrostatic charge, and the greater the charge density is, the smaller the fiber diameter is [37,38]. Ultimately, hierarchical fibers were formed as shown in Fig. 2b.

The appearance of prepared i-SPE-6 is shown in Fig. S2. It can be seen that the surface of i-SPE-6 is smooth and PEO is evenly cast. SEM images of i-SPE-6 are shown in Fig. 3. It can be seen that the PEO matrix is well supported by electrospun nano-fiber networks and trunk fibers act as the scaffolds in the membrane (Fig. 3a). PEO completely permeates into the pores of the electrospun membrane (Fig. 3b and 3c), ensuring the completion of the conductive matrix and enhancement of mechanical strength.

3.2. Crystalline, thermal and mechanical properties of electrospun copolymer and SPEs

WAXS profiles of PVDF-*b*-PPEGMA copolymer, electrospun PVDF-*b*-PPEGMA copolymer (EBCM) and SPEs, and DSC curves of EBCM and SPEs are shown in Fig. 4.

The characteristic peaks of corresponding to α and β-phase crystalline of PVDF are present in the WAXS profile of PVDF-*b*-PPEGMA copolymer [39], as shown in Fig. 4a. Typically, α-phase crystalline can be transferred to β-phase crystalline in PVDF under an electrical force [40]. Therefore, a distinct peak at $2\theta = 20.26°$corresponding to the β-phase crystalline of PVDF is observed in WAXS profile of EBCM. Notably, β-phase crystalline of PVDF is observed in all WAXS patterns of

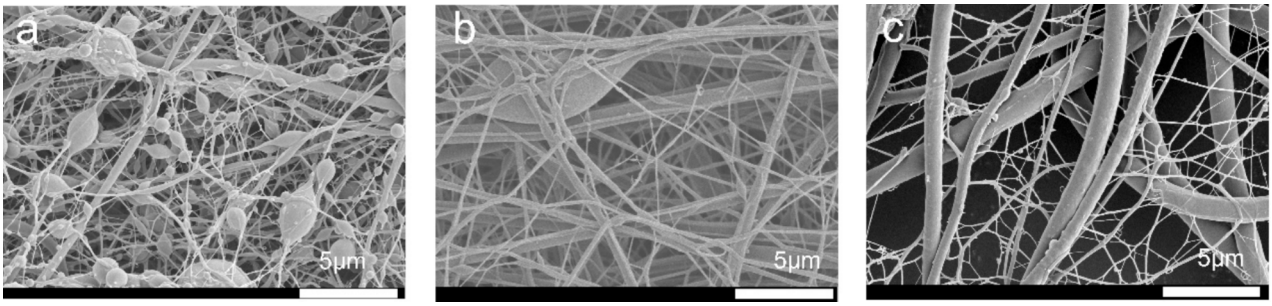

Fig. 1. SEM images of electrospun PVDF-*b*-PPEGMA copolymer fibers with different PVDF/PPEGMA block ratios (a) 100/4.7, (b) 100/7.3, (c) 100/10.

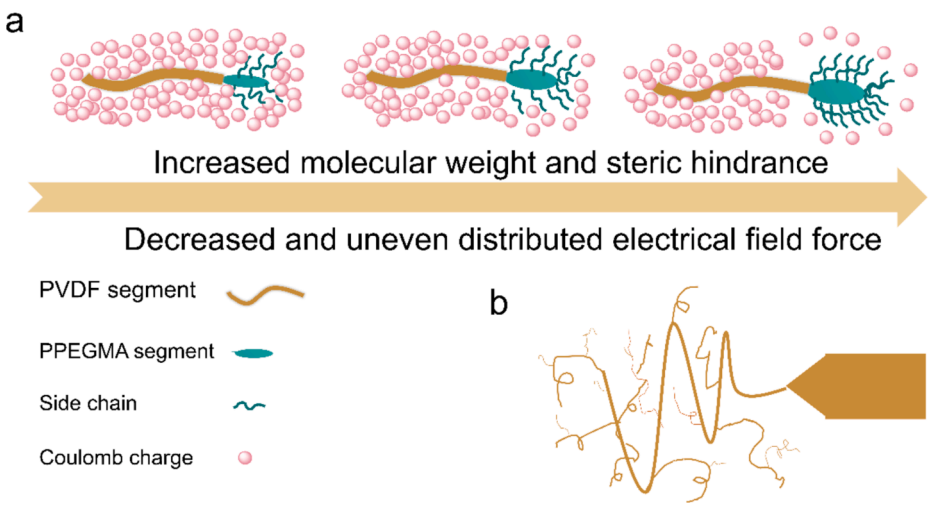

Fig. 2. Schematically illustration of the mechanism of (a) strength and distribution of electrostatic field forces hindered by PPEGMA segment, (b) fibers split during electrospinning process.

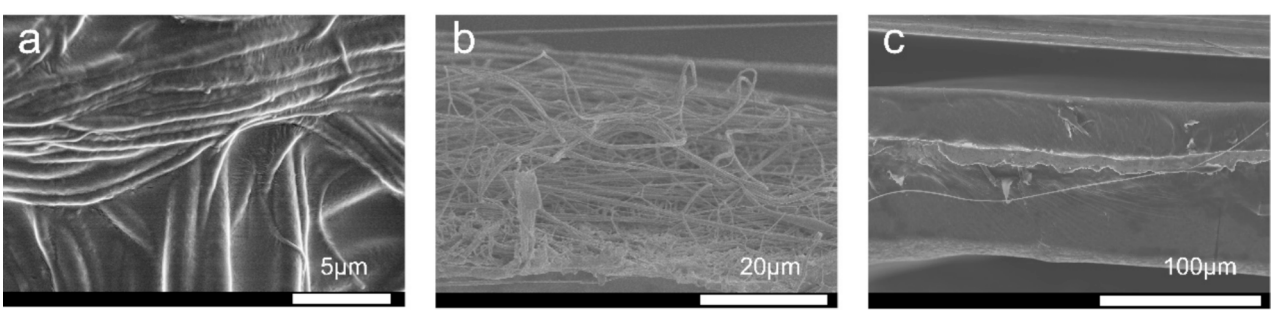

Fig. 3. SEM images of nanofiber network supported PEO/LiTFSi(a), electrospun PVDF-*b*-PPEGMA fibers in fracture surface (b) and fracture surface of i-SPE-6 (c).

SPEs shown in Fig. 4b. On the other hand, the crystallization peak corresponding to PEO does not appear in WAXS patterns of SPEs since the presence of LiTFSi would disrupt the crystallization of PEO.

Electrospun PVDF-*b*-PPEGMA copolymer exhibited a crystallization temperature at 130 °C and a crystalline melting temperature at 171 °C. All SPEs show only one crystallization peak and one crystalline melting peak closed to that of PVDF-*b*-PPEGMA copolymer. The β-phase crystalline structure of PVDF is beneficial to lithium ion transport and amorphous PEO chains exhibit enhanced lithium ion mobility. These distinctive crystallization features of SPEs are instrumental in augmenting the lithium ion conductivity.

Thermal stability is of great importance for the safety and endurability of SPE. TGA curves of SPEs with different EO/Li ratios are shown in Fig. 5a.

All tested SPEs show obvious degradation above 360 °C, and the content of lithium salt in SPE has no significant effect on the thermal degradation behavior. Usually, the liquid polymer electrolyte presented a discouraging thermal stability window. [41] The prepared SPEs are far superior to liquid electrolytes in thermal performance, which can meet application requirements for lithium batteries.

Due to the low molecular weight of PEO, difficulties were encountered in forming stable and continuous PEO films by casting method. The stable and stretchable SPE membranes were obtained using an electrospun PVDF-*b*-PPEGMA copolymer nanofiber membrane as the support. In the typical electrospinning process, fibers exhibit random and inhomogeneous alignments, resulting in loosely stacked structures, particularly at low rotation speeds of the collector drum. Besides, numerous fibers align with the direction of the rotating drum, which imparts anisotropic mechanical properties to the spun membrane. Tensile stress–strain curves of electrospun PVDF-*b*-PPEGMA copolymer

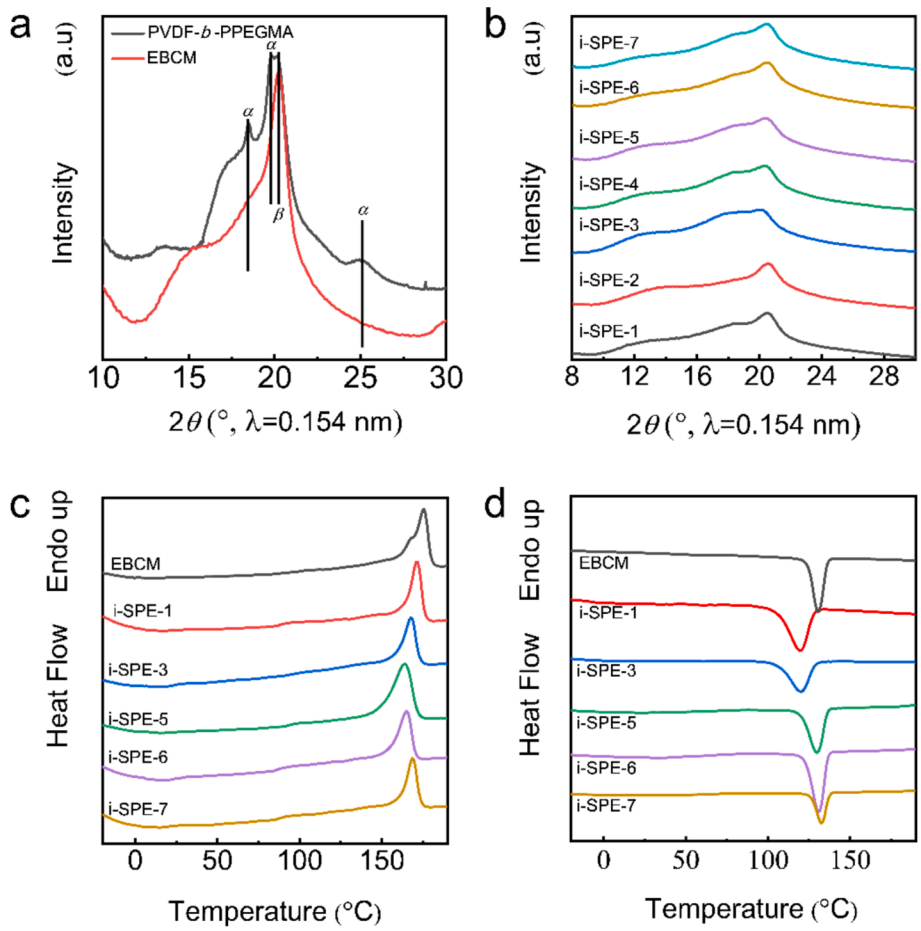

Fig. 4. WAXS profiles of (a) PVDF-*b*-PPEGMA copolymer and EBCM, (b) of SPEs, and DSC curves for EBCM and SPEs (c. second melting process, d. second cooling progress).

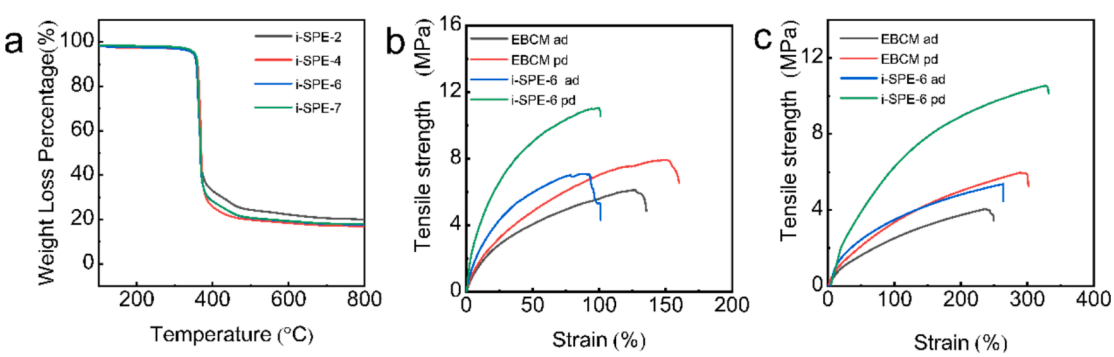

Fig. 5. TGA curves of SPE membranes (a), tensile stress–strain curves of electrospun PVDF-*b*-PPEGMA copolymer membrane (EBCM) and i-SPE-6 membrane at ambient temperature (b) and at 60℃ (c). (ad: along the drum's axis, pd: perpendicular to the drum's axis).

membrane and i-SPE-6 membrane along the drum's axis (denote as ad) and perpendicular to the drum's axis (denote as pd) at ambient temperature and 60 °C are shown in Fig. 5b and 5c, respectively. It can be seen that the tensile strength and the elongation at break of i-SPE-6 membranes are greater in the direction along the rotating drum than that in the direction perpendicular to the drum, and the SPE membrane exhibits greater tensile strengths than the corresponding electrospun block copolymer membrane at room temperature and 60 °C.

The enhancement of mechanical properties of SPE at ambient and high temperatures contributes to the coexistence of hard segment (PVDF) and soft segment (PPEGMA) in the block copolymer and the formation of unique morphology after the electrospinning. The stem fibers act as scaffolds maintaining the strength of the membrane and branch fibers act as network supporting PEO matrix. This structure is beneficial for dispersing the overall stress within the PEO matrix and reinforcing the membrane. The large porous structure of the electrostatic network and the 'adhesion' of the side chain in PPEGMA segment also facilitate penetration of the PEO matrix into pores of the electrospun membrane and firmly 'gripped' by PPEGMA segment. Besides, the content of lithium salt has little effect on the tensile properties of SPE, indicating that electrospun scaffold contributes to maintain the mechanical property for SPE, as shown in Fig. S3b. It can be seen that the mechanical properties of i-SPE-6 are better than that of n-SPE-6 by comparing Fig. 5b, 5c and Fig. S3c. In summary, an ivy-like structure

nanofiber membrane significantly enhances the adhesion between PEO and the nanofiber membrane, enabling the prepared electrolyte to maintain its mechanical stability even under significant tensile forces. The reinforced SPE possesses sufficient strength to suppress the growth of lithium dendrites when in the contact with lithium metal. This, in turn, prevents the electrolyte from being pierced, effectively improving the safety of ASSLMBs.

3.3. Electrochemical performances of SPEs and Li-metal batteries

The electrochemical properties of SPEs are prerequisites and affect their use in rechargeable ASSLMBs. Nyquist curves of the SPEs are shown in Fig. S4. The ionic conductivity of SPEs supported by electrospun PVDF and PVDF-*b*-PPEGMA copolymer membranes at ambient temperature and i-SPE-6 under different temperatures are shown in Fig. 6. It can be seen that the lithium ionic conductivity of SPE supported by ivy-like copolymer fibers is greater than that of SPE supported by electrospun PVDF fibers. The PPEGMA segment exhibits conductivity similar to the PEO matrix. Therefore, the conductivity of the electrospun scaffold comprising PVDF-*b*-PPEGMA copolymer surpasses that of pure PVDF. A significant increase occurs at 60 ℃, corresponding to the melting of PEO. The ionic conductivity at 60 ℃ is 1.05×10^{-4} S·cm^{-1} and one order of magnitude higher than that at a temperature below 60 ℃, which can meet work condition of lithium batteries.

The electrochemical stability window (ESW) is a critical parameter to gauge the stability of an electrolyte. Generally, a wider electrochemical window indicates greater electrochemical stability. As shown in Fig. 6c, the ESW of i-SPE-6 reaches up to 5.58 V (vs. Li^+/Li) at ambient temperature and 5.31 V (vs. Li+/Li) at 60 °C, while the ESW of n-SPE-6 is 5.07 V (vs. Li^+/Li), indicating that the ivy-like structure nanofibers benefit the improvement of the stability of the electrolyte with a Li anode. ESWs of electrospun PVDF-*b*-PPEGMA copolymer and PVDF nanofibers supported PEO SPEs significantly exceed that of the pure PEO electrolyte (4.0 V), which can be attributed to the excellent chemical stability of PVDF. An expanded electrochemical window allows the battery to operate at higher voltages, thereby enhancing the energy density and energy storage capacity within the same volumetric or mass constraints.

Values of I_0, I_{ss}, R_0 and R_{ss} acquired from Fig. 6d, 6e and 6f are listed in Table S1. The calculated lithium ion transference number of i-SPE-6 is 0.56 at ambient temperature and 0.55 at 60 ℃, which is significantly greater than that of PEO-Li electrolyte (0.13) [42]. The lithium-ion transference number of n-SPE-6 is 0.44.

To further investigate the electrochemical performances of as-prepared SPE, i-SPE-6, LFP cathode and lithium metal tablet anode were assembled into sandwich structure cells. The CV, rate and long-term cycling performances of cells at 60℃ are shown in Fig. 7. The charge and discharge curves of cell at 0.2C and 60℃, and under different rates are shown in Fig. S5.

CV result reflects the stability against the Li anode. The first five cycles of CV curves show a pair of redox peaks at 4.0 and 2.7 V without any other peaks for reversible Li stripping/platting processes, as shown in Fig. 7a. Significant overlapped CV curves during the first five cycles in the LFP/i-SPE-6/Li cell reveal its excellent cycling reproducibility, confirming that the SPE are electrochemically stable. Various rates were operated within a range from 0.1 to 2C with every five cycles. As shown in Fig. 7b, i-SPE-6 exhibits the capacity retention of 102, 93.6, 78.9, 61.8 and 33.8mAh g^{-1} at 0.1, 0.2, 0.5, 1 and 2C, respectively. The rate

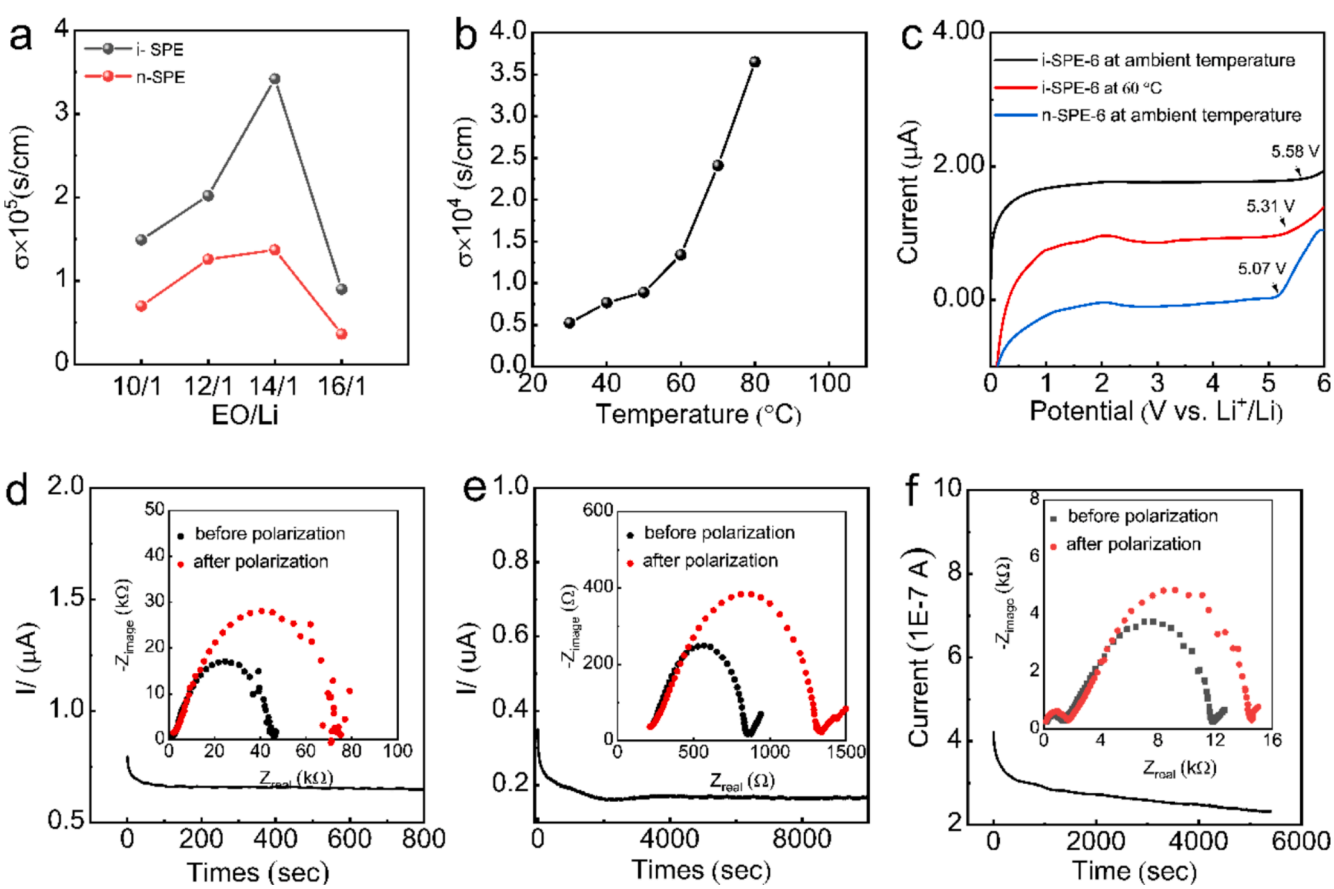

Fig. 6. Electrochemical performances of SPEs: (a) comparison of Li^+ conductivities of SPEs supported by ivy-like copolymer fibers and those supported by normal PVDF fibers, (b) Li^+ conductivities of i-SPE-6 at different temperatures, (c) LSV curves of i-SPE-6 and n-SPE-6, (d) chronoamperometry profile of Li/i-SPE-6/Li symmetric cell at ambient temperature and (e) at 60 °C, (f) chronoamperometry profile of Li/n-SPE-6/Li symmetric cell at 60 °C. Inset: Nyquist impedance spectra of cell before and after polarization.

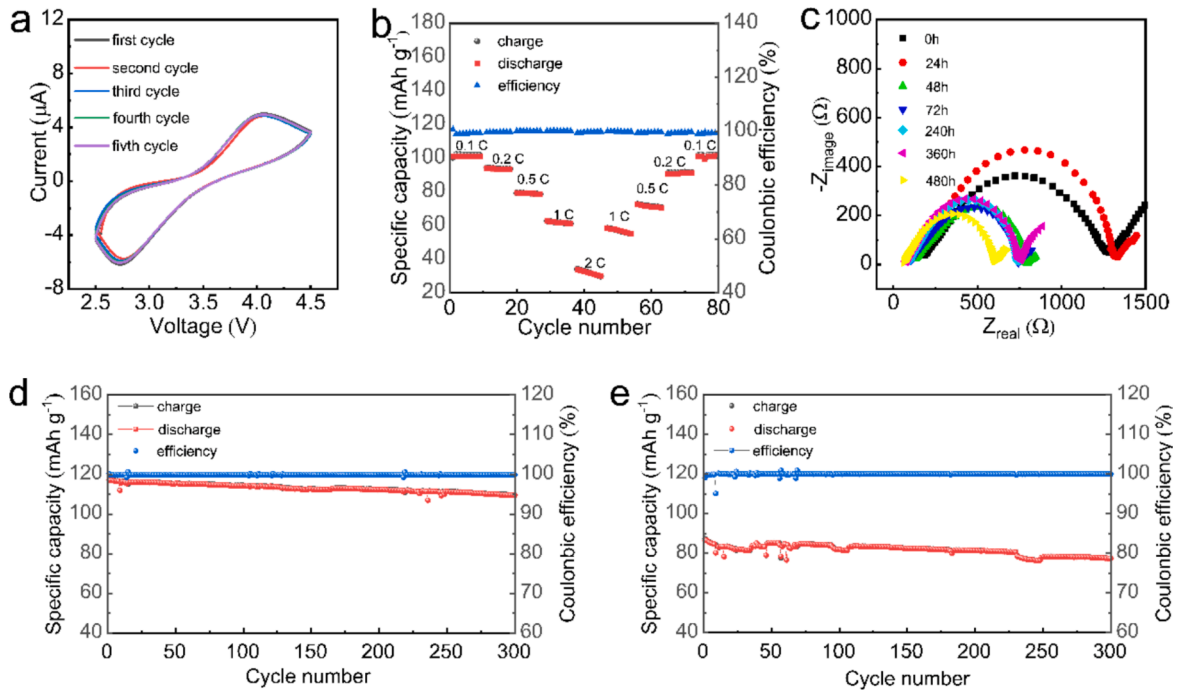

Fig. 7. Electrochemical properties of cells: (a) CV of LFP/i-SPE-6/Li cell at 0.5 mV s^{-1} under 60 °C, (b) rate performance of LFP/i-SPE-6/Li cell at different discharge densities of 0.1, 0.2, 0.5, 1 and 2C, respectively, (c) impedance spectra of the Li/i-SPE-6/Li symmetric cell measured after stored at different times, and cycle charge–discharge plot of LFP/i-SPE-6/Li cell at 0.5C (d) and at 1C (e).

performance is not outstanding, which can be improved by increasing the molecular weight of PVDF and PVDF-*b*-PPEGMA copolymer, and adding of inorganic nanoparticles in the future work. Fig. 7c shows variation of the interface impedance under different storage time at 60 ℃. The initial impendence is 1120 Ω, and the impendence remains basically unchanged after being stored for 24 h. However, the impedance value rapidly decreases to 700 Ω after being stored for 48 h. This is due to the melting of PEO at a high temperature, which reduces the interfacial impedance between SPE and lithium metal. Subsequently, the impedance slightly decreases to 600 Ω with the increase of storage time, and remains stable after being stored for 720 h, indicating the good interface compatibility between SPE and lithium metal. From Fig. 7d, it can be seen that the discharge capacity is 117 mAh g^{-1} initially and slightly decreases to 109 mAh g^{-1} after 300 cycles with a constant Coulombic efficiency of 100 % and a capacity retention rate of 93 % at the discharge/charge rate of 0.5C. When the discharge/charge rate is increased to 1C, the discharge capacity remained 77.5 mAh g^{-1} after 300 cycles with a Coulombic efficiency of 100 % and a capacity retention of 90 %. The capacity fading observed with the increase of the C-rate is attributed to detrimental effects on both the cathode and anode during high-rate charging of lithium batteries, which can ultimately lead to charging capacity decay [43].

To evaluate the stability of i-SPE towards the Li metal anode, a symmetrical cell consisting of Li/i-SPE-6/Li was tested using galvanostatic cycling experiments at 0.1 mA cm^{-2} and 60 °C. During this experiment, lithium was stripped from one electrode and plated onto another through charging and discharging cycles, each lasting 0.5 h. As shown in Fig. 8, the polarization voltage remained steady and was less than 30 mV after 250 h of cycling. This indicates that i-SPE has good stability with lithium metal.

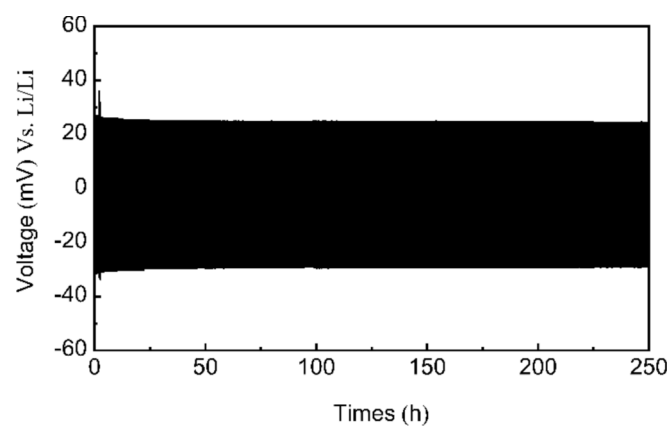

Fig. 8. Galvanostatic cycling curves of the symmetric Li/i-SPE-6/Li cell at current density of 0.1 mA cm^{-2}.

4. Conclusions

A hierarchical-structured nanofiber membrane is fabricated by electrospinning of PVDF-*b*-PPEGMA copolymer and characterized by the presence of interwoven coarse and fine fibers similar to the stems and branches of ivy. The existence of EO units in PPEGMA can effectively improve the compatibility between the block copolymer and PEO molecules. Inspired by the ivy adhesion mechanism, the as-prepared electrospun membrane is scattered among the PEO matrix to prepare a solid electrolyte with good performance. The hierarchical structure and adhesion property enable PVDF-*b*-PPEGMA electrospun membrane to reduce stress and increase the stability of the PEO matrix. Taking the SPE consistent with EO/Li ratio of 14/1 as a sample, the maximum tensile strengths and the elongation at the break reach 10.5 MPa and 333 %. In addition, the incorporation of ivy-characteristic nanofibers enhance the transport rate and uniformity of lithium ions, thereby improving the conductivity and migration frequency of lithium ions. SPE membrane supported by ivy-like nanofibers exhibited a wider electrochemical window than PEO (5.58 V vs 4 V) and excellent cycling stability. The capacity retention reaches 90 % after 300 cycles at 1C. This indicates that the ivy-characteristic electrospun membrane is highly significant in enhancing the electrochemical performance of Li-metal batteries.

CRediT authorship contribution statement

Shanshan Gao: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wenshuo Wang:** Validation, Software, Methodology, Investigation. **Dayang Yu:** Investigation, Formal analysis, Data curation. **Pengju Pan:** Writing – review & editing, Supervision, Formal analysis. **Yongzhong Bao:** Writing – review & editing, Supervision, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Yongzhong Bao reports equipment, drugs, or supplies was provided by Zhejiang University. Yongzhong Bao reports a relationship with Zhejiang University that includes: employment and non-financial support. Pengju Pan reports a relationship with Zhejiang University that includes: employment. Dayang Yu reports a relationship with Zhejiang University that includes: employment. Shanshan Gao reports a relationship with Guizhou University that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.113808.

Data availability

Data will be made available on request.

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